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SYSTEM, METHOD, AND APPARATUS FOR UNDERLID HOME-BASED MEIBOMIAN GLAND DISEASE TREATMENT VIA RESISTIVE HEATING AND MAGNETIC-BASED MECHANICAL STIMULATION

Abstract

A system, method, and apparatus is configured to treat MGD by applying resistive heating and magnetic-based mechanical stimulation of the eyelid. The system includes an underlid device configured to be inserted on the surface of the eye under the eyelid, and a stimulator wand configured to be held by the user on or adjacent to the eyelid in the area of the underlid device. The stimulator wand generates radio frequency (RF) energy that is used to induce an electric current in the underlid device, which is used to excite resistive heating elements that generate heat for heating the meibomian glands. The stimulator wand also generates an oscillating magnetic field that attracts ferromagnetic elements on the underlid device and induces oscillatory movement in the underlid device. The oscillatory movement of the underlid device applies a massaging motion to the meibomian glands while, at the same time, heating the glands via the heating elements.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to a system, method, and apparatus for treating meibomian gland disease/dysfunction (MGD) via resistive heating and magnetic-based mechanical stimulation.

BACKGROUND

[0002] Meibomian gland disease/dysfunction (MGD) is upwards 80% of the root cause issue for patients with dry eye disease. Currently, there are two mildly performing drugs that treat this disorder, another extremely effective method, administered in-clinic by a machine called the LipiFlow, and another older, but also effective method using intense light pulses (ILP), which is also administered in-clinic. Other than the drugs, which have questionable effectiveness, there exists a lack of effective home-based methods for treating MGD dry eye.

[0003] The ocular surface of the eye is coated with a three-layer film commonly referred to as a tear film or tears. Referring to FIG. 1, the three layers of tear film are a mucin layer, which adheres to the ocular surface (epithelium), a middle aqueous layer, and an upper/outer meibum layer. The mucin and aqueous layers are produced by the lacrimal gland and other glands within the conjunctiva of the eye. The meibum layer is produced by meibomian glands in the eyelid. Tears flow from the ocular surface through the nasolacrimal canal, which drains into the back of the nose and throat.

[0004] Referring to FIG. 1, the mucin layer of the tear film is hydrophilic, which helps spread the aqueous layer evenly over the ocular surface. Together, the mucin and aqueous layers aid in maintaining the lubrication across the ocular surface and reducing shear stress during blinking or rubbing. Meibum forms the top layer of the tear film, helps prevent the aqueous layer from evaporating too quickly, which in turn keeps the aqueous layer free of dry spots and thereby lubricates the ocular surface for a longer period of time without needing refreshed. Tear film is anti-microbial and helps maintain the health of the eye, as well as clear any contaminants or particles that might come into contact with the eye. A more detailed cross-sectional illustration of the tear film is shown in FIG. 2.

[0005] Dry eye disease, or DED, affects millions of people worldwide. According to some studies, the third most common reason for visiting an ophthalmologist's office is for dry eye disease symptoms. Recently it has been shown that up to 80% of dry eye cases also have a component called meibomian gland dysfunction or MGD. Normally, the lipid layer produced by the meibomian glands (see FIG. 3) spreads evenly into a thin (thickness in nanometers) protective film over the air-tear interface above the cornea. This lipid can also be referred to as "meibum." Every time a person blinks a slight amount of lipid protective film is spread. There are, however, many conditions under which this oily layer no longer spreads out evenly over the tear film and this process can be interrupted, reduced, or even stopped entirely. These root causes can include but are not limited to hormonal changes in the oil production properties with age, skin mites living in the eyelashes, prolonged infection such as difficult to remove stytes, general inflammation (Blepharitis), autoimmune diseases or allergic reactions, and more recently the inadequate blinking from excessive screen time known as computer vision syndrome (CVS).

[0006] The absence of an outer protective lipid layer increases the evaporation rate of the tear film covering the eye, leading to the possibility of dry spots over the cornea epithelium. This is measured quantitatively as the so-called tear film break-up time metric abbreviated as TBUT or TFBUT.

[0007] There are two main classifications for MGD or dry-eye disease. Aqueous Deficient Dry Eye (ADDE) accounts for approximately 30% of dry-eye disease cases. Evaporative Dry-Eye (EDE) accounts for approximately 60% of dry-eye disease cases.

[0008] With ADDE, the lacrimal gland fails to produce a volume of tears sufficient to effectively coat the eye. This leads to tear hyperosmolarity, a state in which the osmolarity of the tears exceeds that of the epithelial cell, leading to reduced cell volume and increased concentration of solutes.

Factors known to contribute to ADDE include the following:

TABLE-US-00001 Aging Refractive Surgery Low Androgens Contact Lens Usage Autoimmune Disorders Anesthesia Lacrimal Obstructions Frictional Damage

[0009] With EDE, a deficient or unstable tear film lipid layer results in a high rate of tear film evaporation. Factors known to contribute to ADDE include the following:

TABLE-US-00002 Vitamin A Deficiency Contact Lens Usage Preservatives from ADDE Frictional Damage

Drops

[0010] In the past, mild MGD has been addressed by using warm compresses, eyelid cleansing compounds, and massaging the eyelids gently so that the meibum clogging the meibomian gland orifice is expressed or cleared. These approaches, however, have not been shown to clinically be effective in the majority of severe dry eye cases.

[0011] More recent treatments involve heating up the meibomian glands from inside the eyelids to melt or soften the meibum oils clogging the meibomian gland openings. As the eyelids are heated from the inside, heat is delivered directly to the meibomian gland. In some instances, air bladders are also used to massage the meibomian glands to help express the softening oil clogs from the glands. However, such procedures can still be highly invasive and costly procedures requiring an expert ophthalmologist, anesthesia, and multiple treatments each year.

[0012] Other known eye treatments include heating the outside of the eyelids using heating pads. In this type of procedure, ophthalmologists must still use forceps with an intermediate pressure to effectively express the glands. Such treatments are also invasive and can be uncomfortable for the patient. Currently, there are no effective “at home” treatments for MGD.

SUMMARY

[0013] A system, method, and apparatus is configured to treat MGD by applying heat and mechanical stimulation of the eyelid, allowing for self-administration by the user at home in a safe and effective manner.

Description

DESCRIPTION OF DRAWINGS

[0014] FIG. 1 illustrates details of the human eye and the ocular surface of the eye.

[0015] FIG. 2 illustrates details of the aqueous/muco-aqueous layer that coats the eye.

[0016] FIGS. 3A and 3B illustrate details of the lower eyelid and the meibomian glands thereof.

[0017] FIGS. 4A-4D illustrate an example configuration of an underlid portion of a system for treating MGD.

[0018] FIGS. 5A-5C illustrate an example configuration of a stimulator wand portion of the system for treating MGD.

[0019] FIG. 6 is a schematic illustration of an example configuration of the system for treating MGD.

[0020] FIG. 7A illustrates the positioning of components of the system for treating MGD during treatment.

[0021] FIG. 7B illustrates the radiofrequency (RF) and electromagnetic (EM) functionality implemented via the components of the system during treatment.

[0022] FIG. 8 illustrates the re-positioning of the system components during treatment.

DESCRIPTION

[0023] A system, method, and apparatus is configured to treat MGD by applying heat and mechanical stimulation of the eyelid. The system is configured to allow for self-administration by the user at home in a safe and effective manner. The system **10** includes an under the eyelid (“underlid”) component or device **20** configured to be inserted on the surface of the eye under the eyelid, and an external, hand-held component or device **100**, referred to herein as a stimulator wand, configured to be held by the user on or adjacent to the eyelid in the area of the underlid device **20**. The stimulator wand **100** generates radio frequency (RF) energy that is used to induce an electric current in the underlid device **20**, which is used to excite resistive heating elements that generate heat for heating the meibomian glands. The stimulator wand **100** also generates an oscillating magnetic field that attracts ferromagnetic elements on the underlid device **20** and induces oscillatory movement in the underlid device. The oscillatory movement of the underlid device **20** applies a massaging motion to the meibomian glands while, at the same time, heating the glands via the heating elements.

[0024] The underlid device **20** that covers the area of the meibomian glands without covering the cornea. As shown in FIGS. 4A-4D, the underlid device can have a shape configured to correspond to the anatomy of the eyelid and the meibomian glands. While the underlid device **20** is shown in FIGS. 4A-4D as having a generally rectangular configuration, other configurations, such as one having a curved edge or surface for following the contour of the eyelid, can be implemented. Covering only the area of the eyelid necessary to treat the meibomian glands avoids the need for anesthetization of the eye surface, which allows the system to be used for self-administered at-home MGD treatment.

[0025] Referring to FIGS. 4A-4D, the underlid device **20** includes a substrate **22**, such as a flexible printed circuit board (PCB) upon which components are mounted and electrically connected by electrical vias printed/etched on the PCB substrate. The underlid device **20** has a radio frequency (RF) energy harvesting design, including resistive heating elements **30** that are powered via an RF induced electrical current. To facilitate this, the underlid device includes an RF receiving (RX) antenna **24** for energy harvesting. As shown in FIGS. 4A-4D, the RF RX antenna **24** can be a coil that extends around the periphery of the underlid device. The RF RX antenna **24** can, for example, be a wire coil laminated onto the PCB substrate **22**, or can be printed/etched onto the PCB substrate.

[0026] The underlid device **20** includes an integrated circuit (IC) **26** that is electrically connected to the antenna **24** and to the heating elements **30**, e.g., via traces **28**. The IC **26** is configured to rectify power harvested via the antenna **24**, to distribute power to the heating elements **30**, and to control the operation of the heating elements. One or more temperature probes **32** can be distributed on the underlid device **20** and used by the IC **26** to sense the temperature of the device for purposes of safety and control of the heating elements **30**. The IC **26** can, for example, utilize a closed-loop control scheme for maintaining a desired temperature of the underlid device **20** using temperature sensed via the probes **32** as a feedback.

[0027] As shown in the side view of FIG. 4D, the heating elements **30** are positioned on an eyelid side of the underlid device **20**, and are shielded by a thin stack **40** positioned on a sclera side of the device. In the example configuration of FIGS. 4A-4D, there are eight heating elements **30** arranged in groups of two, each of which have an associated thin stack **40**. Thus, the example configuration of the underlid device **20** includes four thin stacks **40**.

[0028] As shown in the zoomed sectional detail of FIG. 4C, each thin stack **40** includes a reflective thermal heat shield **42** positioned adjacent the associated heating element(s). Adjacent to the heat shield **42** (beneath it, as viewed in FIG. 4C) is a ferromagnetic structure **44**, which can be constructed of a permanent magnet material or a ferromagnetic material with a high magnetic permeability. The ferromagnetic structure **44** can, for example, be constructed of a mu-metal

material, which is a class of highly permeable nickel-iron soft ferromagnetic alloys. Adjacent to the ferromagnetic structure **44** (beneath it, as viewed in FIG. **4C**) is a low absorption (k) thermal insulating material **46**, such as an aerogel material. The thermal insulating material **46** can also extend around the ends/edges of the heat shield **42** and ferromagnetic structure **44**. As shown in [0029] FIG. **4C**, the thin stack **40** shields the sclera from the heating elements **30**, which act on the eyelid, specifically the meibomian glands.

[0030] All of the components of the underlid device—the PCB substrate **22**, IC **26**, heating elements **30**, and the thin stack, are encapsulated in a body of overmolded material **50** that serves as a moisture barrier and provides a biocompatible interface with the eye/eyelid. The body **50** can, for example, be constructed of a silicone hydrogel material. As shown in the side view of FIG. **4D**, the body **50** can include recesses or trenches **52** formed around the heating elements **30** on the eyelid side of the device. The recesses/trenches **52**, in effect, raise the position of the heating elements **30** so that they project from the eyelid side of the device. This helps increase the local pressure with which the heating elements **30** are held against the eyelid and meibomian glands and therefore improves heat transfer from the heating elements to the eyelid. Additionally or alternatively, features such as dimples and/or chevrons can also be used to achieve this function.

[0031] The MGD treatment system **10** also includes a handheld stimulator device **100**, referred to herein as a stimulator wand, that is used in conjunction with the underlid device to apply heat and mechanical stimulation to the eyelid. This is shown in FIGS. **5A-5C**. As shown in these Figs., the stimulator wand **100** includes an RF transmitting (TX) antenna **102** for power transmission to the underlid device, and for communication with the underlid device. The stimulator wand **100** also includes an electromagnet **104** which, as an example, includes an electromagnetic (EM) coil **106** wrapped around a ferrite core **108**. The electromagnet **104** is energizable to generate the magnetic field that attracts the ferromagnetic structure **44** in the thin stack(s) **40** of the underlid device **20** to apply pressure to the eyelid. The RF TX antenna **102** and electromagnet **104** are positioned at an application end **110** of the stimulator wand **100**, along with a pressure sensor **112** configured to sense the squeeze pressure applied to the eyelid during use. The end **112** of the stimulator wand **100** that engages the eyelid is formed from or coated with an anti-microbial material.

[0032] The stimulator wand **100** also includes an electronics module **120** and a battery **114** for powering the electronics module. The electronics module **120** can, for example, include an application-specific integrated circuit (ASIC). The electronics module **120** is illustrated schematically in FIG. **6** and includes a processing component **122**, memory devices **124** (RAM/NVRAM), communication radio(s) **126** (e.g., Bluetooth, NFC, Wi-Fi, etc.), and digital, solid-state devices **130** for switching/modulating power to the RF TX antenna **102** in order to induce current in the RF RX antenna **24** of the underlid device **20**. The processor **122** is configured to execute a software application that controls the operation of the stimulator wand **100** and its components. The stimulator application, settings, and other data/applications are stored in the memory **124**. The communication radio **126** is configured to communicate with a connected device **134** running a custom application **136**, which is referred to herein as a smartphone running a custom smartphone app. Other implementations of the connected device **134** and custom application **136** can, of course, be implemented. Examples can include a tablet computer or a PC (desktop or notebook).

[0033] The communication radio **126** can also allow for uploading program data, settings, software/firmware updates, and any other data from the smartphone **134** to the stimulator wand **100**. Stimulation data can include, for example, stimulation wave forms for the RF TX antenna **102** and parameters/algorithms for modulating the DC signal supplied to the EM coil **106**. Similarly, the radio communication can allow for downloading stimulation data from the stimulation wand **100** to the app **136** on the smartphone **134**. The downloaded stimulation data can be used to enable tracking or logging of stimulation therapy through a variety of channels, including: [0034] Recording and storing user inputs. [0035] Recording and storing stimulator settings. [0036] Storing

safety settings and parameters. [0037] Tracking applied pressure. [0038] Tracking wand motion via onboard accelerometer/gyroscope. [0039] Tracking treatment day/date/time. [0040] Tracking treatment schedule and sending reminders. [0041] Tracking treatment frequency and duration. [0042] Querying and recording effectiveness of treatment.

[0043] The smartphone app **136** can utilize internet connectivity to act as an interface between the stimulator wand **100** and the internet/cloud. In doing so, software/firmware updates, stimulator settings, and other data can be downloaded to the smartphone **134** from the internet and then transferred to the stimulator wand **100** from the smartphone. Similarly, stimulator settings, logged stimulator data, and other data can be transferred from the stimulator wand **100** to the smartphone **134**, and uploaded from the smartphone to the internet/cloud. This internet connectivity can, for example, allow for a third party, such as a physician, to adjust stimulator settings and review stimulation logs.

[0044] The electronics module **120** is operatively connected to the RF TX antenna **102** and to the EM coil **106** so that their respective operations can be controlled. A user interface **140**, which can include input buttons **142** and/or knobs, as well as a small screen **144** (LED or LCD, for example). The electronics module **120** controls the operation of the RF antenna **102** and EM coil **106** in response to inputs received via the user interface **140**. All components of the stimulator wand **100** are mounted in or on a housing **150** that can be ergonomically designed for handheld use. For reference, the stimulator wand **100** can have a form factor similar in size to that of a whiteboard marker.

[0045] Through the user interface **140**, the user can control the operation of the stimulator wand **100** according to the stimulation settings stored in memory **124**. RF settings (e.g., amplitude, frequency, etc.) define how the RF signal is controlled to produce a desired heating of the underlid device **20**. EM settings define how the DC pulse applied to the EM coil **106** produces an EM field to induce a desired pulsating/vibratory motion through magnetic coupling to the ferromagnetic structure **44** of the underlid device **20**.

[0046] To track treatment, the stimulator wand **100** can include an instrument **152** accelerometer and/or gyroscope for position, attitude, and motion sensing. These components can also be mounted on the ASIC of the electronics module **120**. Through this treatment tracking, the system **10** can analyze treatment motions and compare them with the therapeutic results to determine which movements are effective and which are not. Treatment tracking can also monitor the amount of time spent at locations along the eyelid and the pressure applied at those locations to determine whether more treatment time should be spent at certain locations. Through this tracking, the system can learn what treatments methods are most effective and can offer instructions or suggestions to the user accordingly.

[0047] At the same time, the system **10** can also track treatment date, time of day, and duration and can send reminders accordingly. The system **10** can also build and maintain a therapy log and use the data stored therein to further refine the treatment regimen. Through this, the system **10** can determine the best thermal stimulation settings for melting the meibum clogging the meibomian glands, and the best mechanical stimulation settings to produce an eyelid squeezing and/or massaging motion for expressing the meibum from the glands.

[0048] As shown in FIGS. 7A and 7B, the stimulator wand **100**, via the RF TX antenna **102** emits an RF signal that excites the RF RX antenna **24** on the underlid device **20**. The RF signal emitted from the stimulator wand is rectified and controlled by the control module **120** to energize the heating elements **30** on the underlid device. As shown in FIG. 7B, the RF signal can be modulated to increase/decrease its strength or amplitude which, in turn, can be used to increase/decrease the amount of heat applied by the underlid device and the resulting temperature of the inner eyelid. The temperature is remotely sensed via the temperature sensors **32** and transmitted to the controller **120** which uses the sensed temperature as feedback for controlling the application of heat to the eyelid. The stimulator wand **100** and the underlid device **20** can therefore cooperate to control heat

application, to limit the amount of heat applied to the eyelid, and to serve as a safety mechanism for preventing overheating.

[0049] At the same time, the electromagnet coil **104** on the stimulator wand **100** is energized, and the coils **106** create a magnetic field that acts on the ferromagnetic core **108** which, in turn, focuses the magnetic field at the end of the stimulator wand. When the stimulator wand **100** is positioned close to the underlid device **20**, the magnetic field acts on the magnetic elements **44** in the thin stacks **40** of the underlid device to magnetically attract the underlid device. This magnetic attraction between the stimulator wand **100** and the underlid device **20** creates a mechanical force that exerts pressure on the eyelid, squeezing the eyelid between the stimulator wand and the underlid device. The pressure sensor **112** of the stimulator wand **100** can be configured to sense the applied squeeze pressure.

[0050] Referring to FIG. 7B, while the eyelid is squeezed, the position of the stimulator wand **100** can be manipulated by the user to cause a corresponding change in position of the underlid device **20**. The user can thus move the underlid device **20** to different positions under the eyelid so that the MGD treatment can be applied over the entire width of the eyelid. The magnetic field itself can also be modulated in order to induce a corresponding movement of the underlid device **20**. The magnetic field can, for example, be modulated to cause the underlid device to apply a pulsatory or vibrational movement to the eyelid.

[0051] Advantageously, the RF TX signal emitted by the antenna **102** of the stimulator wand **100** is generated via an adjustable alternating current (AC) signal, whereas the magnetic field emitted by the electromagnet **104** of the stimulator wand **100** is generated by a switchable (via SS switches **130**) direct current (DC) signal. This allows the two signals emitted by the stimulator wand **100** to coexist without interfering with each other.

[0052] FIG. 8 illustrates the operation of the system **10**, method, and apparatus **20**, **100** to treat MGD by applying resistive heating and magnetic-based mechanical stimulation of the eyelid.

[0053] The underlid device **20** is positioned between the eyelid and the eyeball. The stimulator wand **100** is positioned against the exterior surface of the eyelid, and the EM coil **106** is energized to create a magnetic field that acts on the underlid device **20**, which draws the underlid device toward the stimulator wand, compressing the eyelid between. The pressure sensor **112** at the tip of the wand **100** senses the pressure applied to the tip, which is representative of the squeeze placed on the eyelid. At the same time, the RF TX antenna **102** is energized to induce a current in the RF RX antenna **24** of the underlid device **20** for powering the heating elements **30**. As a result, the eyelid and the meibum in the meibomian glands is heated and squeezed.

[0054] As shown in dashed lines in FIG. 8, movement and other manipulations of the stimulator wand **100** magnetically induces corresponding movement of the underlid device **20**. This allows the location of the pressure applied to the eyelid to be adjusted, as indicated generally by the arrows in FIG. 8. Lateral positional adjustments allow the therapy to be applied across the width of the eyelid. Vertical positional adjustments moves the pressure in the direction of the length of the meibomian gland ducts, which helps expel the meibum from the glands. The MGD treatment regimen proceeds through this combination of movements.

[0055] Additionally, the stimulator wand **100** can be configured to modulate the frequency and/or amplitude of the DC current supplied to the EM coil **106** signal to produce pulsation in the generated magnetic field. This modulation can be configured to induce vibratory movements in the underlid device **20**, which produces a corresponding pulsation in the pressure applied to the eyelid and the meibomian gland. This pulsating pressure helps break-up, dislodge, and move the meibum through the meibomian gland ducts, and to expel the meibum from the meibomian glands.

[0056] Advantageously, the pressure is applied to the eyelid without placing excess pressure on the eyeball. The pressure sensor **112** at the tip of the stimulator wand **100** senses the pressure applied to the eyelid and can modulate the EM coil signal to control the application of pressure within predetermined bounds, for both treatment efficacy and for safety purposes. By distributing the

ferromagnetic material **44** of the thin stacks **40** across the area of the underlid device **20**, the pressure can be applied uniformly across the underlid device, providing both repeatable and reliable MGD treatment.

[0057] The electromagnet **104** includes a ferromagnetic core **108** and an EM coil **106** wrapped around the core. The configuration of the electromagnet **104** and its components implemented in the stimulator wand **100** can differ in form so as to comply with the configuration and form factor of the stimulator wand, without departing or detracting from the features and function described herein.

[0058] The ferromagnetic core **108** can be constructed of a high magnetic permeability material, such as soft iron. The EM coil **106** can be constructed of a relatively thin, e.g., 36 gauge, EM wire. For the above example, assuming a typical 3 mm eyelid thickness, a 1.4 Tesla electromagnet will provide a magnetic squeeze force sufficient to provide the desired pressure between the stimulator wand and the underlid device. To produce this, in an example configuration, the soft iron forming the core can have a relative magnetic permeability (μ_r) of 5,000, and the EM coil is formed from 36 gauge EM wire at about 100 turns/inch and 400 ohms/foot resistance.

[0059] According to this design, at **100** turns/inch, a 1-inch long electromagnet with a ¼-inch wide head has a length of approximately:

$$100 \times 2\pi \times \frac{1}{4}\text{-inch} \approx 157 \text{ inches} \approx 13 \text{ feet.}$$

Exciting the EM coil having this configuration at a coil current of 0.2 A leads to a voltage drop of 1 volt and produces an approximately 2.5T magnetic field.

[0060] Additionally, the shape of the magnet core **108** produces a magnetic field where the EM field coupling and EM field gradient creates lateral forces on the underlid device to lock its lateral position in place. This can be produced, for example, by configuring the core with converging legs around which the EM coil is wrapped. This configuration shapes the flux field of the electromagnet **104** in a manner that captures and constrains the underlid device. This creates stability in the underlid device **20**, as it is constrained from moving out of alignment with the tip of the stimulator wand **100**. This is especially beneficial since the intent is to move the wand **100** and to produce a magnetically induced movement of the underlid device **20** while, at the same time, exerting a squeeze pressure on the eyelid.

[0061] The following outlines an example implementation of the MGD treatment system **10** as used to treat MGD: [0062] First, the underlid device **20** is inserted in the underlid region of the upper or lower eyelid, as the system **10** can be used to treat MGD in either location. [0063] Next, the tip of the stimulator wand **100** is brought close to the outside surface of the eyelid proximate to the underlid region where the underlid device **20** is positioned. [0064] The user then activates the system through the user interface **140** on the stimulator wand **100**, which initiates the transmission of RF energy to apply heat to the lid via the heating elements **30** in the underlid device **20**. The temperature sensors **32** in the underlid device **20** will detect temperature on the eyelid surface (heating surface) as well as the eyeball surface and ensure the operation is within the safety range. [0065] Once the sensed temperature reaches a predetermined threshold, the EM coil **106** is energized with a DC current, causing the electromagnet **104** to “grab” onto the ferromagnetic elements **42** of the underlid device **20** and sandwich the eyelid. EM coupling between the stimulator wand **100** and the underlid device **20** can be confirmed by the pressure sensor **112** in the stimulator wand. [0066] With the underlid device **20** coupled to the electromagnet **104** in the stimulator wand **100**, the EM current can be modulated to produce pulsation in the squeeze force. The pressure sensor **112** on the tip of the smart wand **100** will ensure no excessive force will be applied to the eyelid due to excessive current driving the electromagnet **104**. The pressure sensor **112** can also sense the user inadvertently applying pressure via the manual maneuvering of the stimulator wand **100**. [0067] The user utilizes the magnetic coupling between the stimulator wand **100** and the underlid device **20** to slide the underlid device along the eyelid to mechanically

stimulate meibomian glands in a “wiping” motion for expressing the meibum from the glands. The underlid device **20** can be moved to another treatment area by repositioning the stimulator wand **100**. [0068] Throughout the treatment, utilizing the accelerometer/gyroscope **152** features, the stimulator wand **100** can track the wand motions used to expel the meibum and to keep track of the underlid locations that are treated. The treatment time/duration at each location is recorded, as is the thermal and mechanical characteristics of the applied treatment (wiping pattern, or pulsed action, heat levels, etc.) at each location. All of this recorded data can be transmitted (e.g., via Bluetooth connection) to the external device **134** via the app **136**, which can transmit the data to the cloud/internet for remote access and processing. The app **136** can give the user useful feedback on the areas that have been treated and can coach/instruct on where additional treatment should be focused. This feedback/coaching can come from remote processing on the cloud/internet. [0069] The app **136** can send reminders to the user for follow up treatments, and can also send push notifications to ask the user caring questions related to their dry eye disease and treatment regimen such that the treatment becomes proactive instead of reactive. [0070] The system **10** can also use the collected data, along with data related to the user answers regarding treatment queries, so that the system can learn what types of treatment are most effective. The system **10** can then coach/instruct the user on future or additional treatments that can be implemented according to what are determined to be the most effective treatment routines. [0071] What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the present invention is intended to embrace all such alterations, modifications, and variations that fall within the spirit and scope of the appended claims.

Claims

1. A stimulator system for massaging an eyelid to move meibum through a meibomian gland duct toward a gland duct orifice at the eyelid margin in order to clear meibum blockage in the gland duct, comprising: an underlid device configured to be positioned between the eyelid and the eyeball surface, the underlid device comprising a ferromagnetic element; a stimulator wand comprising an electromagnet configured to produce an electromagnetic (EM) field; wherein the stimulator wand is configured to be positioned against an outer surface of the eyelid, and the electromagnet is configured to be energized so that the EM field acts on the ferromagnetic element in the underlid device to magnetically couple the stimulator wand to the underlid device.
2. The stimulator system recited in claim 1, wherein the underlid device further comprises one or more heating elements and an RF receiver antenna for powering the heating elements via an inducted current generated in the RF receiver antenna, wherein the stimulator wand comprises an RF transmitter antenna configured to generate an RF signal that acts on the RF receiver antenna of the underlid device to induce the current in the RF receiver antenna.
3. The stimulator system recited in claim 2, wherein the EM field produced by the electromagnet is oscillating and induces oscillatory movement in the underlid device configured to apply a massaging motion to the meibomian glands while, at the same time, heating the glands via the heating elements.
4. The stimulator system recited in claim 2, wherein the underlid device further comprises an integrated circuit (IC) that is electrically connected to the RF receiver antenna and to the heating elements, the IC being configured to rectify power harvested via the RF receiver antenna, to distribute power to the heating elements, and to control the operation of the heating elements.
5. The stimulator system recited in claim 2, wherein the underlid device further comprises one or more temperature probes distributed on the underlid device and used by the IC to sense the

temperature of the device for purposes of safety and control of the heating elements.

- 6.** The stimulator system recited in claim 2, wherein the heating element and ferromagnetic element are arranged in a thin stack on the underlid device, wherein each thin stack comprises a reflective thermal heat shield positioned adjacent the heating element and a low absorption (k) thermal insulating material adjacent the ferromagnetic element.
- 7.** The stimulator system recited in claim 2, wherein the underlid device comprises a body of overmold material that encapsulates the underlid device.
- 8.** The stimulator system recited in claim 7, wherein the body of overmold material comprises a silicone hydrogel material.
- 9.** The stimulator system recited in claim 2, wherein the underlid device comprises recesses formed around the heating elements on the eyelid side of the device, the recesses being configured to effectively raise the position of the heating elements so that they project from the eyelid side of the underlid device to increase the local pressure with which the heating elements are held against the eyelid and meibomian glands to improve heat transfer from the heating elements to the eyelid.
- 10.** The stimulator system recited in claim 2, wherein the RF signal can be modulated to increase/decrease its strength or amplitude in order to produce a corresponding increase/decrease in the amount of heat applied by the underlid device.
- 11.** The stimulator system recited in claim 2, wherein the magnetic attraction between the stimulator wand and the underlid device creates a mechanical force that exerts pressure on the eyelid, squeezing the eyelid between the stimulator wand and the underlid device.
- 12.** The stimulator system recited in claim 2, wherein the stimulator wand includes a pressure sensor configured to sense the applied squeeze pressure.
- 13.** The stimulator system recited in claim 2, wherein the stimulator wand is configured to modulate the magnetic field produced by the electromagnet in order to induce a corresponding pulsatory and/or vibrational movement of the magnetically coupled underlid device.
- 14.** The stimulator system recited in claim 2, wherein the RF signal emitted by the stimulator wand is generated via an adjustable alternating current (AC) signal, and the magnetic field emitted by the stimulator wand is generated via a switchable direct current (DC) signal.
- 15.** The stimulator system recited in claim 2, wherein the electromagnet comprises electromagnetic (EM) coil wrapped around a ferrite core.
- 16.** The stimulator system recited in claim 2, wherein the stimulator wand includes an electronics module comprising an application-specific integrated circuit (ASIC) that includes one or processors, memory devices, communication radios, and digital devices for switching/modulating power to the RF transmitting antenna and electromagnet.
- 17.** The stimulator system recited in claim 16, wherein the processor is configured to execute a software application that controls the operation of the stimulator wand and its components, the stimulator application, settings, and other data/applications being stored in the memory, and the communication radio is configured to communicate with a connected device running a custom application.
- 18.** The stimulator system recited in claim 16, wherein the communication radio is configured to upload program data, settings, software/firmware updates, and any other data from the connected device to the stimulator wand.
- 19.** The stimulator system recited in claim 16, wherein the communication radio is configured to downloading stimulation data from the stimulation wand to the connected device.
- 20.** The stimulator system recited in claim 19, wherein the downloaded data can include user inputs, stimulator settings safety settings and parameters, applied pressure, wand motion, treatment day/date/time, treatment schedule and reminders, treatment frequency and duration, and querying and recording effectiveness of treatment.
- 21.** The stimulator system recited in claim 17, wherein the connected device is configured to utilize internet connectivity to act as an interface between the stimulator wand and the internet, wherein

the internet connectivity enables software/firmware updates, stimulator settings, and other data to be downloaded to the connected device from the internet and then uploaded to the stimulator wand from the connected, and wherein the internet connectivity enables stimulator settings, logged stimulator data, and other data to be downloaded from the stimulator wand to the connected device, and uploaded from the connected device to the internet.

22. The stimulator system recited in claim 17, wherein the connected device comprises a smartphone.

23. The stimulator system recited in claim 2, wherein the stimulator wand comprises an onboard accelerometer and/or gyroscope for tracking the position, attitude, and motion of the stimulator wand.

24. The stimulator system recited in claim 16, wherein the stimulator wand comprises a user interface to the electronics module, wherein the electronics module controls the operation of the RF antenna and EM coil in response to inputs received via the user interface.

25. The stimulator system recited in claim 2, wherein the electromagnet comprises a magnet core is configured to produce a magnetic field in which the electromagnetic field coupling and electromagnetic field gradient creates lateral forces on the underlid device to constrain movement of the underlid device.

26. A method for treating meibomian gland dysfunction (MGD), comprising the steps of: providing the stimulator system recited of claim 2; inserting the underlid device in the underlid region of the upper or lower eyelid; positioning the tip of the stimulator wand close to the outside surface of the eyelid proximate to the underlid region where the underlid device is positioned; activating the stimulator wand to initiate the transmission of RF energy to the heating elements to apply heat to the eyelid via the underlid device; energizing the EM coil so that the electromagnet attracts the underlid device so that the stimulator wand and the underlid device are magnetically coupled and apply a squeeze pressure to the eyelid between the underlid device and the stimulator wand; and modulating the EM current to produce pulsation in the squeeze pressure.

27. The method recited in claim 26, further comprising the step of utilizing the magnetic coupling between the stimulator wand and the underlid device to slide the underlid device along the eyelid to mechanically stimulate meibomian glands in a “wiping” motion for expressing the meibum from the glands and/or to move the underlid device to another treatment area by repositioning the stimulator wand.
